

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-aaae59c2f22e41e20.jsonl`  
- SHA-256: `a7934146fc5ed0a625a936d660b5521352a02cdf416ba0016fb1b57eb0417d19`  
- Source modified: `2026-06-16T03:20:18+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:19:16.226Z)

Source Extractor

Research question: "Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMM/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,

edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Fetch and extract key claims from this source:

****URL:**** <https://arxiv.org/abs/2103.06669>

****Title:**** Temporal Action Segmentation from Timestamp Supervision (Li et al., CVPR 2021)

****Found via:**** Small-data / weak / timestamp-supervised TAL + active learning search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-16T03:19:18.379Z)

I'll fetch the source and extract key claims.

3. user (2026-06-16T03:19:29.553Z)

Paper Summary: Temporal Action Segmentation from Timestamp Supervision

****Title:**** Temporal Action Segmentation from Timestamp Supervision

****Authors:**** Zhe Li, Yazan Abu Farha, Juergen Gall

****Venue:** CVPR 2021**

****Publication Date:** Submitted March 11, 2021; revised March 26, 2021**

Abstract

The authors propose using timestamp supervision for temporal action segmentation, claiming it requires "comparable annotation effort to weakly supervised approaches" while providing stronger supervisory signal than ordered action lists alone.

Concrete Claims Extracted

(1) Annotation Cost Reduction

The paper states: "Timestamps require a comparable annotation effort to weakly supervised approaches, and yet provide a more supervisory signal." However, ****no quantitative reduction metrics are provided**** comparing annotation time/cost between full frame-wise labeling, timestamp supervision, and weakly supervised methods.

(2) Proposed Method

Two specific technical contributions are described:

- ****Action change detection:**** "Our approach uses the model output and the annotated timestamps to generate frame-wise labels by detecting the action changes."
- ****Confidence loss:**** "We further introduce a confidence loss that forces the predicted probabilities to monotonically decrease as the distance to the timestamps increases."

(3) Performance Results

****No specific accuracy numbers or quantitative comparisons are provided in this abstract.**** The paper only claims: "models trained with timestamps annotations achieve comparable performance to the fully supervised approaches" without citing numerical results.

(4) Datasets

The abstract mentions: "The evaluation on four datasets" but ****does not name them**** in the content provided.

4. assistant (2026-06-16T03:19:32.193Z)

The abstract-level fetch lacks the specific numbers (datasets, accuracy). Let me get the deeper version of the paper for falsifiable numbers.

5. user (2026-06-16T03:19:40.130Z)

maxLength size of 10485760 exceeded

6. assistant (2026-06-16T03:19:42.249Z)

The PDF is too large. Let me try the HTML/ar5iv version for the specific numbers.

7. user (2026-06-16T03:19:45.186Z)

REDIRECT DETECTED: The URL redirects to a different host.

Original URL: <https://ar5iv.org/abs/2103.06669>

Redirect URL: <https://ar5iv.labs.arxiv.org/abs/2103.06669>

Status: 301 Moved Permanently

To complete your request, I need to fetch content from the redirected URL. Please use WebFetch again with these parameters:

- url: "https://ar5iv.labs.arxiv.org/abs/2103.06669"
- prompt: "Extract: (1) the names of the four datasets used for evaluation, (2) reported segmentation accuracy / F1@{10,25,50} / edit numbers comparing timestamp-supervised models to

fully-supervised baselines and to weakly-supervised baselines ? give exact percentages, (3) how many timestamps are annotated per video/segment (e.g., one timestamp per action segment), (4) any statement about annotation time reduction comparing timestamp annotation to full frame-wise labeling (e.g., seconds per video). Quote exact sentences and numbers."

8. user (2026-06-16T03:19:55.646Z)

Temporal Action Segmentation from Timestamp Supervision - Key Findings

(1) Four Datasets for Evaluation

The study evaluated on: **50Salads, Breakfast, BE0ID, and Georgia Tech Egocentric Activities (GTEA)**.

(2) Accuracy/F1/Edit Comparisons

****Timestamp-Supervised vs. Fully-Supervised (50Salads):****

- Timestamp approach: 75.6% accuracy, F1@50: 60.1%, Edit: 66.8
- Full supervision: 77.8% accuracy, F1@50: 58.6%, Edit: 63.8

****Timestamp-Supervised vs. Transcript-Supervised (50Salads):****

- Timestamp: 75.6% accuracy
- Transcript methods: 54.7% (CDFL), 49.4% (NN-Viterbi), 45.5% (HMM-RNN)

****Breakfast Dataset:****

- Timestamp: 64.1% accuracy, F1@50: 47.4%, Edit: 69.9
- Full supervision: 68.0% accuracy, F1@50: 51.5%, Edit: 69.4

(3) Timestamps Per Video/Segment

"Only one frame is annotated from each action segment" in the training videos.

(4) Annotation Time Reduction

"Annotators need 6 times longer to annotate the start and end frame compared to annotating a single timestamp." The paper emphasizes that timestamp annotation requires only pressing "a button when an action occurs" rather than "going slowly back and forth between the frames."

9. assistant (2026-06-16T03:19:58.220Z)

I now have the specific, falsifiable numbers. I have enough verified detail to extract the claims.

10. user (2026-06-16T03:20:08.885Z)

Structured output provided successfully

11. assistant (2026-06-16T03:20:18.711Z)

I've extracted and verified the key claims from the Li et al. (CVPR 2021) paper on Temporal Action Segmentation from Timestamp Supervision.

Source quality: primary (peer-reviewed CVPR 2021 paper, authors Zhe Li, Yazan Abu Farha, Juergen Gall).

The five claims, all backed by direct quotes pulled from the ar5iv full-text (the abstract-only fetch lacked numbers, so I went to the HTML rendering for the specific figures):

1. (central) Timestamp supervision reaches near-full-supervision quality ? 50Salads 75.6% acc / F1@50 60.1 / Edit 66.8 vs full 77.8% / 58.6 / 63.8, actually beating full supervision on F1 and Edit.
2. (central) ~6x cheaper annotation than marking start/end boundaries.

3. (central) The mechanism: model-output-driven action-change detection between timestamps + a monotonic confidence loss ? this directly produces boundaries.
4. (supporting) Timestamps beat transcript/ordered-list supervision by ~20+ points on 50Salads.
5. (supporting) Validated on four datasets (50Salads, Breakfast, BE01D, GTEA), with Breakfast reproducing the near-full result.

Relevance to the research question: highly applicable to Dimension 3 (small-data) and Dimension 1 (windowing/boundaries). The confidence loss directly attacks the project's over-merge failure mode (it forces probabilities to decay away from each rally's timestamp, encouraging crisp boundaries between bridged rallies), and the 6x annotation saving plus near-full-supervision parity is a strong fit for a ~6?16-video golden set where one click per rally is far cheaper than marking exact rally start/end frames. License note: I could not verify the paper's reference implementation license from this fetch ? that needs a separate check before any code reuse.